Many choices are possible for the conservative force field [219, 244, 342]. In case the DPD equations are used simply to control the momentum (and not as a coarse-graining device), any empirical molecular force field may be used. When DPD is used for coarse-graining, it is common to adopt a simplified pair potential, namely

$$\boldsymbol{F}_{ij}^{\mathrm{C}} = \begin{cases} \alpha\left(1 - \dfrac{r_{ij}}{r_c}\right)\hat{\boldsymbol{q}}_{ij}, & r_{ij} < r_c; \\ 0, & r_{ij} \geq r_c, \end{cases} \tag{8.82}$$

where r_{ij} is the interparticle distance, $\hat{\boldsymbol{q}}_{ij} = r_{ij}^{-1}(\boldsymbol{q}_i - \boldsymbol{q}_j)$, parameters α define the strength of the repulsion of particles i and j, and r_c is the cutoff radius. The use of a cutoff is important for efficiency reasons—in a large system, only the particles interacting at close range will generate nonzero forces of interaction so the force calculation may be substantially reduced. Both α and r_c could be allowed to be different for different particle pairs, although they are most often taken to be homogeneous. The pair potential corresponding to the force is

$$\varphi_{\mathrm{DPD}}(r) = \begin{cases} \dfrac{\alpha r_c}{2}\left(1 - \dfrac{r}{r_c}\right)^2, & r < r_c; \\ 0, & r \geq r_c. \end{cases} \tag{8.83}$$

so that the conservative potential energy is:

$$U_{\mathrm{DPD}} = \sum_{i<j} \varphi_{\mathrm{DPD}}(r_{ij}), \qquad r_{ij} = \|\boldsymbol{q}_i - \boldsymbol{q}_j\|. \tag{8.84}$$

The choice of dissipative and random forces is made so as to maintain a prescribed temperature. The typical formulas used for this purpose are

$$\boldsymbol{F}_{ij}^{\mathrm{D}} = -\gamma\omega^{\mathrm{D}}(r_{ij})(\hat{\boldsymbol{q}}_{ij} \cdot \boldsymbol{v}_{ij})\hat{\boldsymbol{q}}_{ij}, \tag{8.85}$$

$$\boldsymbol{F}_{ij}^{\mathrm{R}} = \sigma\omega^{\mathrm{R}}(r_{ij})\theta_{ij}\hat{\boldsymbol{q}}_{ij}, \tag{8.86}$$

where γ and σ parameterize the forces (again, they could be taken to be different for different pairs), and ω^{D} and ω^{R} are configuration-dependent weight functions that typically incorporate a cut-off similar to that used in the conservative force, e.g. one may take

$$\omega^{\mathrm{R}}(r) = \begin{cases} 1 - \dfrac{r}{r_c}, & r < r_c; \\ 0, & r \geq r_c, \end{cases} \tag{8.87}$$

where r_c is the same cutoff used in the conservative force. The relative velocities are denoted by $\boldsymbol{v}_{ij} = \boldsymbol{p}_i/m_i - \boldsymbol{p}_j/m_j$, and θ_{ij} is a symmetric ($\theta_{ij} = \theta_{ji}$) Gaussian

white-noise term with the following stochastic property

$$\langle \theta_{ij}(t) \rangle = 0, \qquad \langle \theta_{ij}(t)\theta_{kl}(t') \rangle = (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})\delta(t - t'), \tag{8.88}$$

and are independent for each pair of interacting particles at each time step. Invariance of the canonical distribution under these equations of motion follows if

$$\omega^{\mathrm{D}}(r_{ij}) = [\omega^{\mathrm{R}}(r_{ij})]^2, \quad \sigma^2 = 2\gamma \mathrm{k_B T}, \tag{8.89}$$

where $\mathrm{k_B}$ is the Boltzmann constant and T the equilibrium temperature. Note that the "canonical distribution" in the case of the DPD system or other momentum-conserving thermostat should have a density which is reduced to the leaves of constant total momentum, i.e.

$$\hat{\rho}_\beta \stackrel{\mathrm{def}}{=} Z^{-1} e^{-\beta H(\boldsymbol{q},\boldsymbol{p})} \times \delta \left[\sum_{i=1}^{\mathrm{N}} p_{i,x} - \pi_x \right] \delta \left[\sum_{i=1}^{\mathrm{N}} p_{i,y} - \pi_y \right] \delta \left[\sum_{i=1}^{\mathrm{N}} p_{i,z} - \pi_z \right]$$

where $\boldsymbol{\pi} = (\pi_x, \pi_y, \pi_z)$ is the total momentum vector.

The conservation of the total momentum is a direct result of the symmetric form of the various forces, thus DPD may be viewed as an isotropic Galilean-invariant thermostat which preserves hydrodynamics [8]. The ergodicity of DPD (i.e. that $\hat{\rho}_\beta$ is the unique measure sampled by the dynamics) has only been demonstrated in the case of a high density of particles in one dimension [334].

The DPD system may be viewed as a special case of the generalized Langevin dynamics with equations

$$\begin{aligned} \mathrm{d}\boldsymbol{q} &= \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t, \\ \mathrm{d}\boldsymbol{p} &= -\nabla U(\boldsymbol{q})\mathrm{d}t - \boldsymbol{\Delta}(\boldsymbol{q})\boldsymbol{p}\mathrm{d}t + \boldsymbol{\Sigma}(\boldsymbol{q})\mathrm{d}\mathbf{W}, \end{aligned} \tag{8.90}$$

where the potential U is given in (8.84), and the damping matrix $\boldsymbol{\Delta}$ and stochastic multiplier matrix $\boldsymbol{\Sigma}$ are related by

$$\boldsymbol{\Delta} = \frac{\boldsymbol{M}^{-1}}{2\mathrm{k_B T}} \boldsymbol{\Sigma}\boldsymbol{\Sigma}^T.$$

In the next subsection we describe the numerical treatment of the DPD system. We then take up the alternative scheme, the pairwise Nosé-Hoover-Langevin (PNHL) method, of [228] which also conserves momentum exactly. Other alternatives to the DPD system include the Peters thermostat [299], the Lowe-Andersen method [241], and the Nosé-Hoover-Lowe-Andersen thermostat [347]. We do not discuss these schemes here, but note that they are compared to DPD and some of the other schemes introduced below in [228].